HUD HACKATHON · RL FOR HARDWARE

FizzVolt

The EDA toolchain, as a reward function.

RL for verifiable, manufacturable circuit design.

One chip. Ten exact connections.

DATASHEET Step-down regulator — typical application

1. `C_IN` ≥ 10 µF within 2 mm of VIN→GND
2. Bootstrap cap `C_BST` 0.1 µF, SW→BST
3. Divider sets `V_OUT = 0.8·(1+R1/R2)`
4. Catch Schottky SW→GND, **cathode to SW**
5. Inductor on SW; `C_OUT` on VOUT→GND
6. EN pull-up to VIN; short FB trace

~10

PARTS PER IC

1

WRONG PART = A RE-SPIN

DAYS–WEEKS LOST

Looks right. Doesn’t build.

- ✕ `D1.1/D1.2` — LED pins are **A / K**
- ✕ USB-C `CC2` floating — **won’t enumerate**
- ✕ MCU `VDD` — **0 decoupling caps**
- ✕ Buck — **no inductor, no catch diode**

0 / 5

RAW BUCK DESIGNS BUILT

Plausible ≠ buildable.

The toolchain becomes the reward.

Prompt → DSL → Real KiCad

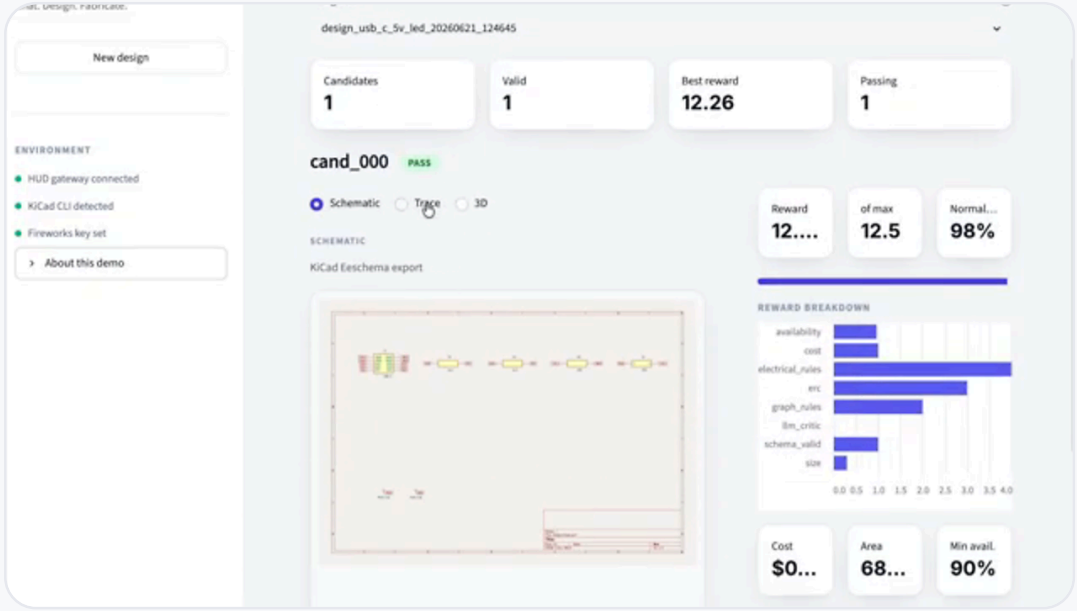
ERC + validators → Reward 0–1

Deterministic compile

ERC + validators

One 0–1 reward

31 tasks



ERC: 0 ERRORS

REWARD 12.26 / 12.5

GRPO turns verification into improvement.

Reward only what passes

Within-group variance = gradient

Graded by real KiCad

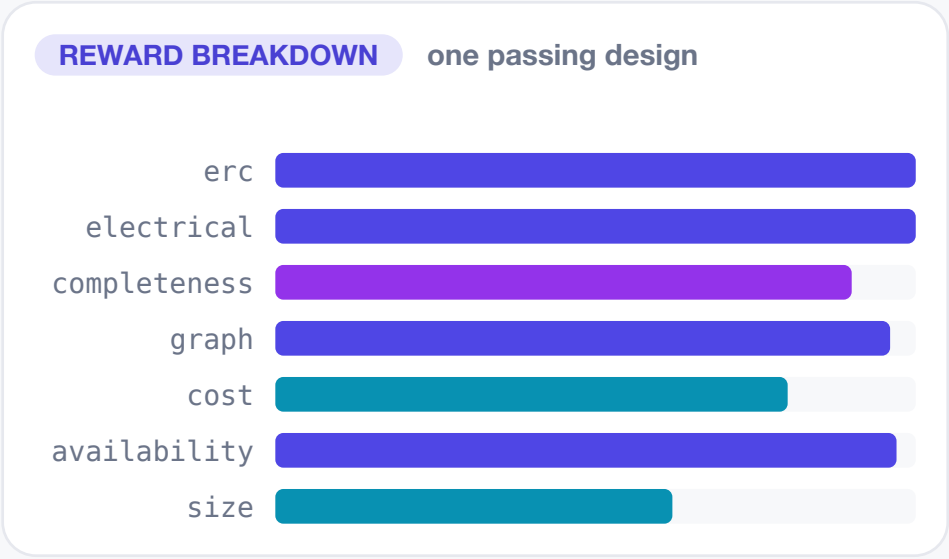
Qwen3-8B on HUD

8

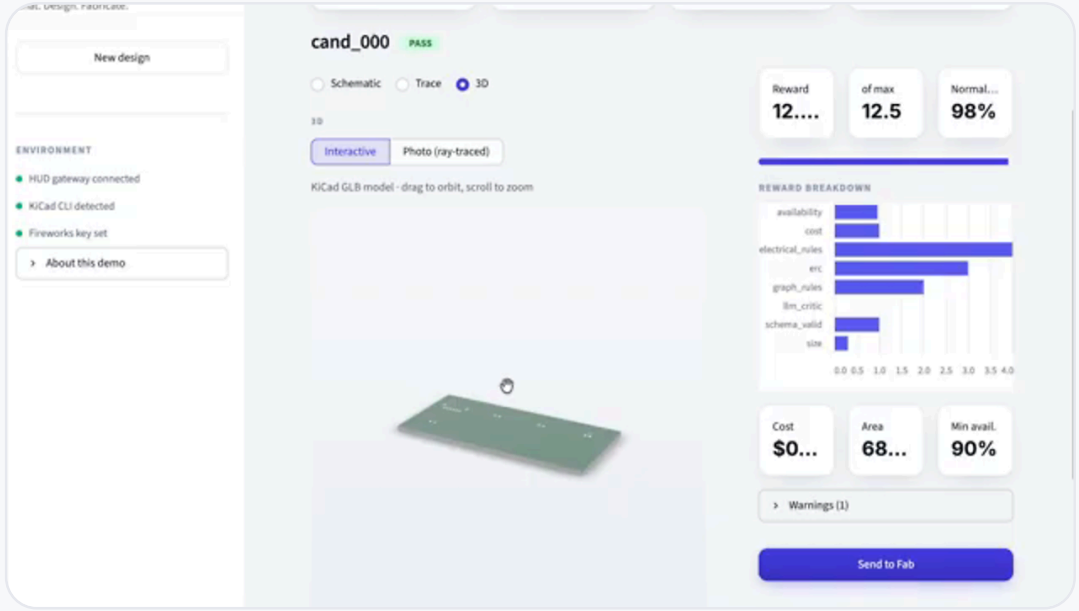
REWARD DIMENSIONS

98%

OF MAX ON PASSES



A sentence → a manufacturable board.



SCHEMATIC → TRACE → 3D → SEND TO FAB

Fewer re-spins

Validated automation

Next: PCB layout & routing

Don’t just look right — build.